

AVAYA INFINITY MCP CONSOLE

Solutions Consultant Technical Demo & Presentation

A technical deep-dive into why I built this application, what it shows under the hood, and how it directly demonstrates the business value of Avaya's actual Infinity roadmap.

Presented by: Jeff Edwards

Savannah, GA | 678-481-5135 | sbfall96@gmail.com | [linkedin.com/in/sbfall96](https://www.linkedin.com/in/sbfall96) | github.com/sbfall96-chx/avaya-mcp-demo

Why I Built This Application: SC Perspective

Bridging Abstract Architecture with Quantifiable C-Suite Value

The Engineering & Demo Imperative

- **To 'Show, Not Just Tell':** Standardized protocol frameworks like Model Context Protocol (MCP) and Lakehouses are abstract concepts. I built this console to make them visual, interactive, and immediately understandable to C-level buyers.
- **Sales Enablement Focused:** Created a client-ready playground that functions offline (perfect for situations with spotty network coverage) or connects directly to live AI APIs for real tool-calling validation.
- **Realistic Scenario Sandbox:** Includes pre-configured incident presets (wait-time spikes, churn escalations) that sales engineering teams can run in real-time to walk clients through day-in-the-life contact center challenges.

Interview Alignment & Technical Rigor

- ✓ **Demonstrating SC Competencies:** Proves my ability to architect a full-stack dashboard, model complex database nodes, implement visual analytics (Chart.js), and code responsive CSS layouts.
- ✓ **Strategic Solution Strategy:** Directly showcases my grasp of Avaya's product portfolio—highlighting the value proposition of AXP cloud telemetry and ACM core media layers.
- ✓ **Interactive Value Pitch:** Designed specifically to address Carlos and other sales leaders' target needs—focusing on ROI calculators, compliance controls, and SLA penalty mitigations.

What the Application Actually Shows

An Interactive Guided Tour of the Core Modules Built

1. Live Protocol Trace (The 'How')

Provides step-by-step transparency. It captures raw JSON inputs, tool schemas, and output payloads exchanged between the model client and MCP database servers, demystifying AI decision logic.

3. Interactive Network Playground (The 'Why')

Animates active connection lines and flowing data packets. Instantly models legacy integration mess ($N \times M$ spaghetti) vs the clean unified bus ($N + M$ MCP) and Edify policy routing layouts.

2. Unified Lakehouse & Silos (The 'What')

Visualizes simulated CRM profiles, SQL database logs, and SharePoint Knowledge Base tables. Proves that models can retrieve siloed customer data without copying or duplicating data repositories.

4. Databricks Governance Log (The 'Security')

Logs compliance activities in real-time. Highlights PII redactions, RBAC access blocks, and logs all gateway actions directly into Unity Catalog audit tables for total transparency.

How It Represents Avaya's Actual Infinity

Translating Architecture Guidelines to Interactive Solutions

Core Infrastructure Pillars

- **AI-Agnostic BYOAI Framework:** In the UI, the operator can switch AI models seamlessly. This models Avaya's commitment to avoiding model lock-in, letting customers bring any LLM client to their data.
- **Zero-Copy Data Access:** Under the hood, the console queries database tables dynamically via schema tool definitions. Data remains in-place, aligned with Avaya's zero-copy architecture roadmap.
- **Edify Journey Flowchart:** An animated flowchart diagram mapping dynamic CX stream packet routes, illustrating how Avaya Infinity handles no-code policy orchestration.

Operational Controls & Telemetry

- ✓ **Tandem Care Checkpoint:** Triggers a visual checkpoint requiring supervisor authorization for high-stakes actions, reflecting Avaya's human-in-the-loop safety roadmap.
- ✓ **Multi-Layer Telemetry Coexistence:** Visualizes cloud events (AXP) and voice trunk parameters (ACM Jitter/Loss/MOS) side-by-side. Demonstrates Avaya's capability to bridge legacy telephony cores with cloud operations.
- ✓ **Databricks Unity Catalog Alignment:** Incorporates PII HIPAA masking and RBAC query block triggers, proving enterprise-grade compliance safeguards.

Detailed Walkthrough: Unified Lakehouse

Silo Consolidation & Interactive Data Management Features

Why Silos Are in the Console

- **Simulating Real-World Friction:** Enterprises have customer history, ticket CRM data, and policies separated across different database platforms. This view shows them all in one place.
- **Zero-Copy Verification:** Rather than migrating databases to the LLM, the model issues on-the-fly read queries to separate tables, illustrating zero-copy architectural advantages.
- **Interactive SharePoint Document Viewer:** Users can select actual warranty, tax, or outage documents from a visual index table and immediately read the text clauses used by the AI model.

Dynamic Data Silos Features

- ✓ **Unified Tables Index:** Includes PostgreSQL Call Database, Salesforce/Zendesk CRM Records, and SharePoint Document Libraries.
- ✓ **Interactive Past Interactions Logger:** Let's the presenter manually write and submit a supervisor log entry back to the customer's CRM history, showcasing active writebacks.
- ✓ **Dynamic Customer Seeding:** Adding a new CRM customer profile automatically seeds historical interaction records based on account tiers (VIP vs Basic).

Detailed Walkthrough: Security & Governance

Centralized Policy Enforcement, HIPAA Protections, and Real-Time Auditing

Governance Controls & Enforcements

- **Dynamic PII Masking:** Automatically redacts sensitive names (e.g. Jeff Edwards -> J*** E*****) in raw database tables, JSON parameters, trace payloads, and final chat outputs.
- **Centralized Proxy Redaction:** Demonstrates that data is protected at the API proxy layer, eliminating the risk of LLM prompts leaking sensitive information.
- **Restricted DB Access Gatekeeping:** Toggling the restriction blocks SQL execution instantly, demonstrating Unity Catalog role-based checks yielding 403 Access Denied messages in the trace.

Governance Audit Dashboard Features

- ✓ **PII Masked Counter:** Running tracker tallying every redaction event occurred in active simulation pipelines.
- ✓ **Blocked Requests Counter:** Real-time count of blocked unauthorized calls, highlighting active security gatekeeping.
- ✓ **Compliance SLA Status Indicator:** Automatically shows 100% SLA Health, which shifts dynamically if unauthorized query violations exceed safe operating boundaries.

Proving the Business Outcome & ROI Case

Translating Telemetry Metrics into Financial Outcomes for Sales Leadership

The Architecture Playground ROI Calculator

- **Quantified Cost Reduction:** Dynamic calculator card showing Legacy custom connectors (\$150,000/yr TCO) vs Avaya Infinity MCP Bus (\$15,000/yr TCO)—a 90% maintenance cost savings.
- **Agility Metric:** Compares 4 months deployment cycles for point-to-point custom adapters against 2 hours for standardized MCP client-server registrations.
- **Security Policy Scope:** Measures the compliance exposure difference between Legacy (High Risk endpoint leaks) and Avaya Infinity (Protected unified gateway).

The Aura AI Business Outcome Banners

- ✓ **Queue Performance (Preset 1):** Dynamic breakout rescheduling and automated chat diversions prevent \$3,200/day in SLA fines.
- ✓ **Contract Security (Preset 2):** Priority routing VIP callers during jitter spikes retains \$24,000 in VIP account contract ARR.
- ✓ **Revenue Retention (Preset 5):** Agentic flow coupon routing saves Jeff Edwards' \$8,400 contract value via SOP matching, proving a 6.6x ROI on retention.

Summary: A Complete Sales Pitch Delivery

Aligning Customer Success, Security Compliance, and IT Budgets

Why I Am Prepared to Stand as an Avaya Solutions Consultant:

1. **Ability to Simplify Complexity:** Translated dry protocol specifications (MCP schemas, JSON payloads) into a high-fidelity visual experience that any buyer can easily understand.
2. **Outcomes-Driven Demos:** Put financial results and TCO metrics front and center, ensuring that the technology is directly coupled with the client's business goals (\$ savings, SLA compliance, ARR retention).
3. **Mastery of the Portfolio:** Handcrafted an application representing Avaya's core architectural tenets—proving I have the technical depth to consult clients on actual roadmaps.